

A.1. a. $(\underline{a} \times \underline{b}) \cdot (\underline{c} \times \underline{d}) \underline{a} : \underline{a}, \underline{b}, \underline{c}, \underline{d} \in \mathbb{R}^3$

$\hookrightarrow (\underline{a} \times \underline{b})_i (\underline{c} \times \underline{d})_i a_n$

$= \epsilon_{ijk} a_j b_k \epsilon_{ilm} c_l d_m a_n$

$= \epsilon_{ijk} \epsilon_{ilm} a_j b_k c_l d_m a_n$

$\epsilon_{ijk} \epsilon_{ilm} = \delta_{jl} \delta_{km} - \delta_{jm} \delta_{kl}$ by identity

$= (\delta_{jl} \delta_{km} - \delta_{jm} \delta_{kl}) a_j b_k c_l d_m a_n$

$= a_j b_k c_j d_k a_n - a_j b_k c_k d_j a_n$

$= (\underline{a} \cdot \underline{c})(\underline{b} \cdot \underline{d}) \underline{a} - (\underline{a} \cdot \underline{d})(\underline{b} \cdot \underline{c}) \underline{a}$

$= \underline{a} [(\underline{a} \cdot \underline{c})(\underline{b} \cdot \underline{d}) - (\underline{a} \cdot \underline{d})(\underline{b} \cdot \underline{c})]$

b.i. $\left(\frac{\delta T_i}{\delta x^k} \right)' = \frac{\delta T'_i}{\delta x'^k} : T'_i = L_{in} T_n$ by tensor law

$= \frac{\delta}{\delta x'^k} (L_{in} T_n)$ by Sub.

$= \frac{\delta L_{in}}{\delta x'^k} T_n + \frac{\delta T_n}{\delta x'^k} L_{in}$ by prod. rule

$= \frac{\delta^2 x^n}{\delta x^i \delta x^k} T_n + L_{kn} \frac{\delta T_n}{\delta x'^k} L_{in}$ by Chain rule & diff.

$= \frac{\delta^2 x^n}{\delta x^i \delta x^k} T_n + L_{in} L_{kn} \frac{\delta T_n}{\delta x'^k}$

ii. $\frac{\delta T_i}{\delta x^k}$ has two free indices, hence would need to satisfy a rank two transformation rule: $L_{in} L_{kn} \frac{\delta T_n}{\delta x'^k}$.

$\hookrightarrow \text{Part (i)} \neq L_{in} L_{kn} \frac{\delta T_n}{\delta x'^k}$

$\therefore$ not a tensor. $\square$

iii. $A'_{ik} = \frac{\delta T'_i}{\delta x'^k} - \frac{\delta T'_k}{\delta x'^i}$

$= \left(\frac{\delta^2 x^n}{\delta x^i \delta x^k} T_n + L_{in} L_{kn} \frac{\delta T_n}{\delta x'^k} \right) - \left(\frac{\delta^2 x^n}{\delta x^k \delta x^i} T_n + L_{kn} L_{in} \frac{\delta T_n}{\delta x'^i} \right)$ by part (i)

$= L_{in} L_{kn} \frac{\delta T_n}{\delta x'^k} - L_{kn} L_{in} \frac{\delta T_n}{\delta x'^i}$

$= L_{in} L_{kn} \frac{\delta T_n}{\delta x'^k} - L_{kn} L_{in} \frac{\delta T_n}{\delta x'^i}$

$= L_{in} L_{kn} \left(\frac{\delta T_n}{\delta x'^k} - \frac{\delta T_n}{\delta x'^i} \right)$ by def.

$= L_{in} L_{kn} A_{nn}$

$\hookrightarrow$ Satisfies rank two tensor transformation rule. $\square$